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RESEARCH INTERESTS

Humanoid Robotics · Wearable Inertial Sensing · Human-to-Robot Motion Retargeting · Safety-Constrained Robot Control · Sim-to-Real Systems

EDUCATION

Chongqing University — *B.Eng. in Robotics Engineering, Mingyue Innovation Class* *Sep 2023 – Jun 2027 (expected)*
College of National Excellence Engineers · **GPA: 3.70/4.0** · Ranked 5/41 in Robotics Engineering (6/58 in Mingyue cohort, Top 10%) · CET-6: 570, CET-4: 598
Selected courses: Microcircuit Design (99), Fundamentals of Robotics (97), Mathematical Physics Methods (97), Automatic Control Principles (92).

RESEARCH EXPERIENCE

Wearable-IMU Upper-Limb Teleoperation & Safety-Constrained Control — *Unitree G1 humanoid robot · advised by Prof. Can Pu* *Nov 2025 – Present*
An end-to-end pipeline that maps a human operator's upper-body motion to the Unitree G1 humanoid via a custom wearable inertial-sensing suit, with closed-form retargeting and configuration-dependent safety constraints validated in simulation and on the real robot.

- **Wearable sensing & fusion.** Designed a 5-node wireless IMU suit (custom ESP32-C3 + ICM-20948 9-axis modules sampled at 225 Hz; self-designed main-control and sensor PCBs) streaming poses to an edge solver over WiFi/UDP, with each link reconstructed relative to the trunk via chest-relative quaternion differencing.
- **Calibration.** Developed sensor-to-segment calibration, including three-pose Procrustes alignment with an axial roll-axis refinement and a one-pose lateral-raise routine; closed-loop calibration residuals reach < 0.0001 rad.
- **Closed-form IK.** Derived an analytical orientation-to-joint solver (YXZ Euler decomposition for shoulder pitch/roll, twist-on-Y projection for the elbow), achieving < 0.001° round-trip error and serving > 80% of frames, with a numerical L-BFGS-B fallback.
- **Safety constraints.** Built an offline URDF / PyBullet self-collision scan over a 120×75 shoulder pitch–roll grid to derive a configuration-dependent yaw feasible region that static joint limits cannot capture; fixed a convex-hull "polarization" artifact (torso mesh inflation) using concave meshes + VHACD decomposition, cutting false collisions from 39.5% to 28.6%. Online limits apply bilinear-interpolated bounds with a conservative margin and a 500 Hz receiver PD state-machine with rate-limited takeover and timeout-to-zero.
- **Sim-to-real.** Tuned the 14-DOF upper-body PD controller via Gaussian-process Bayesian optimization in Isaac Lab (448 trials), reducing simulated shoulder-yaw RMSE from 0.44° to 0.09°.
- **Hardware evaluation & error analysis.** On the real G1, measured 3.43° static and 11.38° dynamic mean joint RMSE and identified shoulder yaw as the dominant bottleneck (17.9× static-to-dynamic error growth). Cross-correlation showed amplitude ratios of 0.98–1.01 across all joints but phase lags of 129–194 ms — indicating the residual error is delay-dominated rather than a mapping-gain deficiency, which redirected subsequent work toward latency reduction.

Output: a full technical report, real-robot and simulation demos, and an open-source firmware package.

Vision-Guided Aerial Delivery System — *National Innovation Training Program, rated Excellent · advised by Prof. Chengliang Wang* *Jul 2024 – Mar 2025*
A two-stage visual guidance system that delivers fire-suppression payloads beyond the ~5–8 m reach and accuracy of manual throwing — a friction-wheel launcher performs coarse aim (industrial camera + NUC) and a guided V-tail aircraft performs fine, in-flight guidance — engineered and evaluated as a repeatable, statistics-driven study rather than one-off tuning.

- **Problem framing & spec.** Quantified the manual-disposal gap — throwing capped at ~5–8 m, accuracy gating the extinguishing effect — and set measurable targets: ≥ 20 m range, ±40 cm / 90 % hit rate, fully-automatic tracking.
- **Two-stage guidance architecture.** Co-designed a hierarchical visual scheme that decomposes the problem into two stages: Stage-1 coarse aim (NUC + 12 mm industrial camera detects the target and corrects launcher yaw and friction-wheel speed) hands off to Stage-2 fine guidance (onboard micro-camera + V-tail control for terminal micro-correction).
- **Avionics & flight control.** Designed and integrated the aircraft hardware stack (STM32H7 main-control + power-management + vision-host + 9-axis IMU + servo-driver boards) as modular, clearly-interfaced subsystems, and implemented V-tail control mixing, IMU attitude estimation, and an outer-loop PID controller; the modular split enabled 5 airframe revisions (duct → plate wing → dual-servo mixing) with little re-debug.
- **Reproducible field evaluation.** Designed and ran a controlled A/B study — guided (two-stage) vs. non-guided (single-stage) aircraft on the same 25 m target, 6 groups × 4 shots each, open field, against a simulated-flame target 50 cm beyond the aim line — logging every impact to score accuracy (mean target-ring score) and dispersion (standard deviation).

- **Quantified result & error analysis.** Two-stage guidance lifted the within-40 cm hit rate above 80 % and the mean score to 5.24 (+42.3 % vs. baseline), at the cost of a 33.7 % rise in landing variance — a statistically-grounded finding that the controller trades dispersion for accuracy and is disturbance-sensitive, redirecting follow-on work to the control law and structural consistency.

Output: rated Excellent at the national level; the launcher + aircraft were later fielded as a RoboMaster dart system (3 target hits across 4 launches, no unrecoverable airframe damage).

SELECTED ENGINEERING EXPERIENCE

RoboMaster 2025 Super Combat — *Software & hardware lead, "Dart" robot* *Nov 2023 – Jul 2025*

- National Third Prize. Built an STM32 state machine with CAN-bus motor control and serial referee communication; FFT-based sensor filtering for stable data; NUC + industrial-camera classical vision for target-panel detection.

Smart Bartending System — *Technical co-founder* *Sep 2024 – Mar 2026*

- Independently developed generations 2–4 of an automated multi-recipe dispensing system. Built the ESP32 + FreeRTOS modular multi-task architecture and the PCB main-control / driver boards; implemented the communication stack (serial HMI, barcode scan, CRC) and a thread-safe task scheduler supporting precise dispensing of 11 recipes.

Low-Cost Smart Sweeping Robot — *Team lead (3 members)* *Nov 2024 – Jan 2025*

- Architected a wireless onboard / remote split (ESP32-S3 edge + remote ROS 2 host running SLAM and path planning) to offload compute and reduce cost by 30% across two iterations.

AWARDS

RoboMaster 2025 Super Combat — National Third Prize	*2025*
National College Student Innovation Training Program — Excellent completion	*2025*
RoboMaster Regional Contest — Second Prize	*2025*
National Undergraduate Electronics Design Contest — University First Prize	*2024*
China Engineering Practice & Innovation Competition — University Bronze	*2024*

TECHNICAL SKILLS

Languages: Python, C/C++, embedded C.

Robotics & simulation: ROS 2, PyBullet, Isaac Lab, Unitree SDK, URDF; analytical & numerical IK, PD/PID control, VQF inertial fusion, Bayesian optimization.

Embedded & hardware: ESP32 / ESP-IDF / FreeRTOS, STM32 (CAN, state machines), PCB design (EasyEDA), NUC + industrial-camera vision, FFT-based signal filtering.